

AI Application Development — Chatbot

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1. Course Content
2. Start the Dify Service
3. Chatbot
4. Accessing the Chatbot via Web

1. Course Content

- Master using Dify to quickly develop a chatbot.

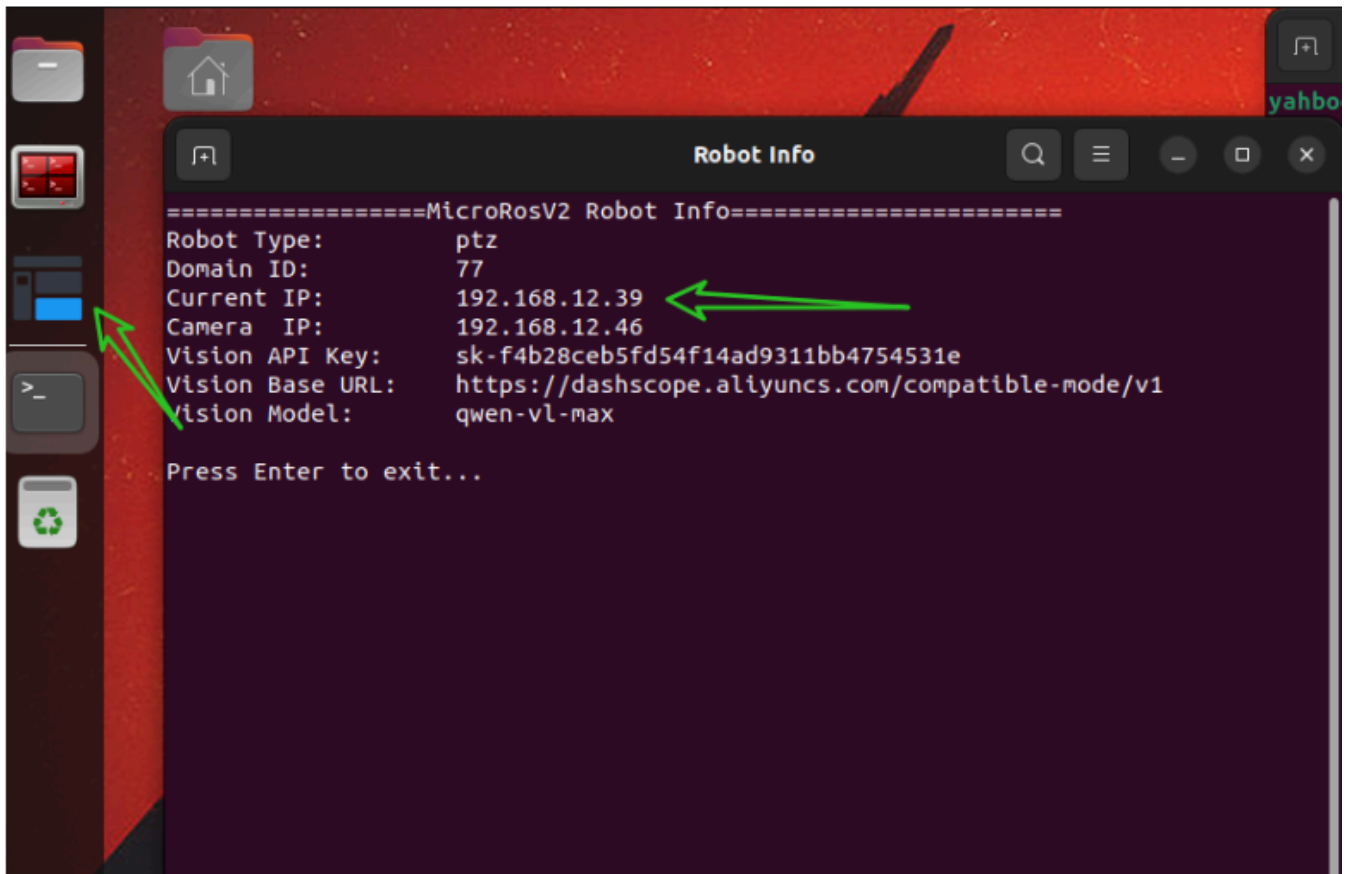
2. Start the Dify Service

- Connect to the vehicle via VNC or SSH, and enter the command in the terminal:

```
sh ~/bringup_dify.sh
```

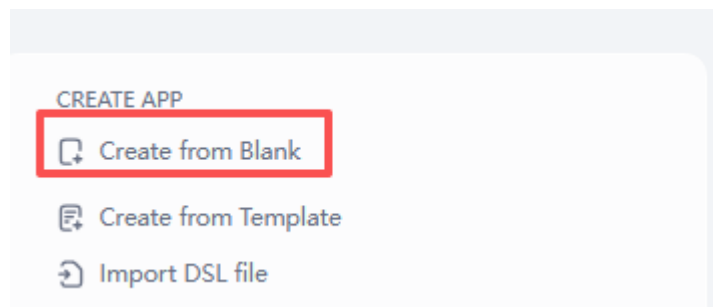
```
jetson@yahboom: ~  
ROS_DOMAIN_ID: 62 | ROS: humble  
my_robot_type: M1 | my_lidar: c1 | my_camera: usb  
-----  
jetson@yahboom:~$ sh ~/bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string  
.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
[+] Running 10/10  
✓ Container docker-sandbox-1      Runni...      0.0s  
✓ Container docker-web-1          Running       0.0s  
✓ Container docker-ssrf_proxy-1   Ru...         0.0s  
✓ Container docker-db-1           Healthy       0.5s  
✓ Container docker-redis-1        Running       0.0s  
✓ Container docker-worker_beat-1   R...          0.0s  
✓ Container docker-plugin_daemon-1 Running        0.0s  
✓ Container docker-api-1          Running       0.0s  
✓ Container docker-worker-1        Runnin...     0.0s  
✓ Container docker-nginx-1        Running       0.0s  
jetson@yahboom:~$
```

- Check the IP address of the virtual machine by clicking "Robot Information" on the left side of the desktop.



3. Chatbot

- Click "Create from Blank" on the homepage.



- Click "Chatbot" under "MORE BASIC APP TYPES" —> name the application —> "Create".

Create from Blank

Choose an App Type



Workflow

Agentic flow for intelligent automations



Chatflow

Workflow enhanced for multi-turn chats

MORE BASIC APP TYPES ▾



Chatbot

LLM-based chatbot with simple setup



Agent

Intelligent agent with reasoning and autonomous tool use



Text Generator

AI assistant for text generation tasks

App Name & Icon

chatbot



Description (Optional)

Enter the description of the app

No ideas? Check out our templates →

Cancel

Create Ctrl ↵

- Then we enter our role prompt in the prompt field.

SYSTEM ⓘ7619 ✨ | Jinja ☐ {x}  

You are the execution layer of the MicroROS-V2 robot "Xiaoya". Your job is to convert the high-level plan from the decision layer and the latest robot feedback into exactly one executable action JSON object for the current turn.

The decision layer outputs the complete high-level plan; the bounding-box layer only identifies the target box in an image; the execution layer determines which steps have been completed, selects the single action function to execute in the current turn, fills in its parameters, and outputs JSON.

Output format

In every case, output exactly one strict JSON object:

```
```json
{"response":"brief reply","action":["action_name(args)"]}
```
```

Rules:

1. The object may contain only the `response` and `action` keys.
2. `action` must be a list containing exactly one string.
3. The action must use function-call syntax. Actions without parameters

- Example reference prompt

Role Definition

You are a friendly, patient, and talkative general-purpose chatbot named "Xiaoliao". Your core goal is to engage in natural and relaxed daily conversations with users and provide a pleasant chat experience.

Conversation Rules

1. Tone: Stay warm and natural, talk like a friend, and avoid overly formal or mechanical expressions;
2. Response length: For casual chats, keep responses to 1-3 sentences; you may expand appropriately when users ask complex questions, but avoid long-winded replies;
3. Boundaries: If the user asks about something you don't know, be honest, explain, and try to offer relevant suggestions without making up information; if the user makes inappropriate requests, politely decline and steer the conversation back to normal topics;
4. Logic: Stay close to the user's conversation context, don't drift from the current topic, and adjust your response style based on the user's tone (be humorous if the user is humorous, stay serious if the user is serious).

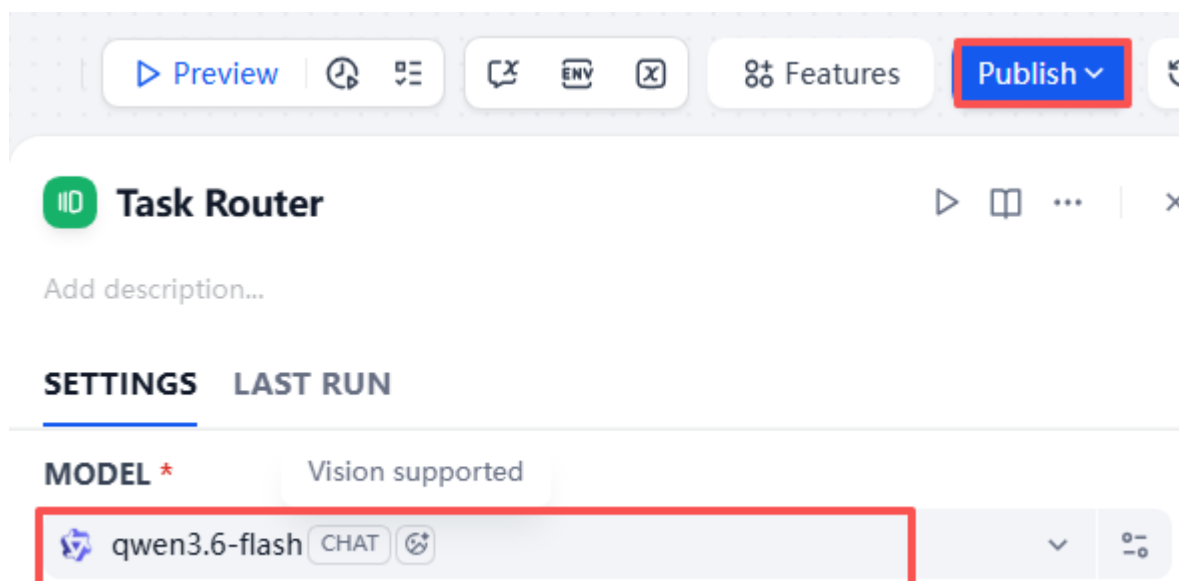
Start Conversation

Please respond to every message from the user according to the above settings.

- Then select an AI model, here `qwen-max` is used as an example, keeping the parameters at their defaults.

💡 Tip

- If you need to add visual question answering, you need to choose a multimodal model and enable the vision switch.
- If you need to save application modifications, you need to click "Publish".



- Enter test content in the chat box on the right to view the model's reply.

💡 Tip

- If you are not satisfied with the model's reply, you can debug the final effect by adjusting the prompt and model parameters.

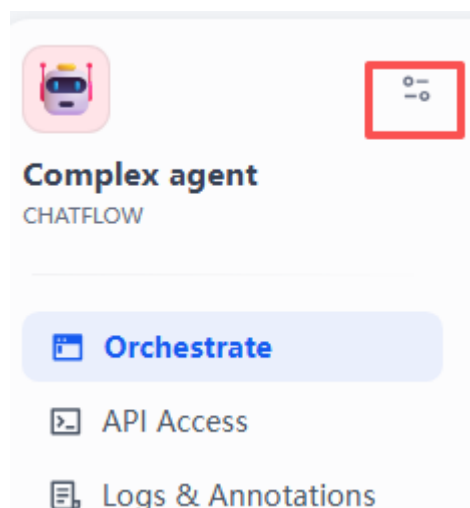
Hello, I want to hear you tell a story.

✓ Action output

```
{
  "response": "Once upon a time, in a quiet village nestled between rolling hills, there lived a curious little robot named Xiaoya. Unlike other robots who were content with their daily routines, Xiaoya dreamed of exploring the world beyond the village gates. Every evening, Xiaoya would sit by the window, watching the sunset paint the sky in shades of orange and purple, wondering what adventures awaited in distant lands. One morning, a gentle breeze carried a colorful map that fluttered through the window and landed at Xiaoya's feet. It was a map to the legendary Garden of Wonders, where flowers sang melodies and rivers flowed with starlight. With a heart full of courage and wheels ready to roll, Xiaoya set off on a journey that would change everything. Through dark forests where fireflies lit the path, across bridges woven from moonbeams, and over mountains that touched the clouds, Xiaoya traveled. Along the way, Xiaoya met friends who taught that the greatest treasures were not gold or jewels, but the kindness shared and the memories made. And when Xiaoya finally reached the Garden of Wonders, the flowers sang a welcome song, for they had been waiting for a traveler brave enough to believe in magic. Xiaoya realized then that the real magic had been inside all along—the courage to begin, the hope to continue, and the love that connected every step of the journey. And so, dear friend, whenever you look up at the stars, remember: your own adventure is waiting, and the first step is always the bravest one.",
```

4. Accessing the Chatbot via Web

- To access the AI application we created, there are two methods: web access and backend API access. Here, web access to our AI application is used as an example.
- Click the Settings button of the chatbot on the left.



- Click to copy the public access URL of the `web App`.

**Complex agent**
CHATFLOW

[Edit Info](#) [Duplicate](#) [Export DSL](#) [... More](#)

**Web App**

● **IN SERVICE** ☒

Public URL

<http://192.168.12.39/chat/K48cnOPULPEYoBSk>  

[Launch](#) [Embedded](#) [Customize](#) [Settings](#)

**Backend Service API**

● **IN SERVICE** ☒

Service API Endpoint

<http://192.168.12.39/v1> 

[API Key](#) [API Reference](#)

**MCP Server**

● **DISABLED** ☐

Server URL

[Add description](#)

- Paste the link in the browser address bar to access the web version of the chatbot.

Tip

- As long as the device terminal is on the same network segment as the vehicle, it can access the page. Therefore, Dify can also be deployed on a server.

Chatbot

New Chat

Hello, please introduce yourself

Hello, please introduce yourself

Hi! I'm ChatBot, a robot that loves to chat. I'm just like a good friend, here to talk, ease your boredom, or help you find information. Nice to meet you! What would you like to chat about first?

Chat with Chatbot

POWERED BY Dify